

CI7521 / 2025-2026 – Assignment 1: Classical Machine Learning

Overview

This group assignment counts for 50% of the overall mark for this module. It includes both group and individual submissions as well as self-assessment. Marking will also consider how the group members rate the contributions of each other. You should form groups of up to four students. Its subject is implementing Classical Machine Learning solutions in Python using the Scikit-Learn library and other libraries introduced in the class. Specifically, classification methods should be applied to the OpenML dataset: GesturePhaseSegmentationProcessed / ID: 4538:

[OpenML: GesturePhaseSegmentationProcessed - ID4538](#)

Deliverables and Submission

The coursework must be submitted by the deadline given on Canvas. Follow the submission guidelines in Canvas. For each submission ensure that you include

- (Group Submission) **A single IPython notebook (.ipynb)** containing all code, text (including references) and results (see Part I below). Any results presented should be directly reproducible from the code without any modification.
- (Group Submission) The IPython notebook must be converted to a **word file** which should also be submitted. Ensure that all code, text and results are included in the word file. If generative AI tools were used for producing the code, an appropriate Appendix must be added at the end of the document. Guidance on how to convert ipynb to docx is given on Canvas.
- (Individual Submission) Report with a specific structure (see below in Part II) in **word or pdf format** with the student's name, k-number and group at the beginning of the document.

Rules

- If you have re-used large chunks of code from third-party sources, clearly specify the chunk (e.g. comment in the relevant cell), cite the source(s) and add references at the end of the notebook.
- Use of generative AI tools is permissible, but you should fully acknowledge it, as described in the University guidance at [AI and Assessment: Digital Learning & Tools](#). In addition, you are expected to provide relevant information in the appendices with a full record of prompts and responses you have received from generative AI tools for each submission (group/individual).
- Usage of any third-party libraries that have not been used in the class must be explicitly approved by the Lecturer by email or mentioned in the coursework's FAQ (available on Canvas).
- In case the above rules are not obeyed, marks may be deducted and/or the submission may be considered for plagiarism and penalised according to the University regulations.

Project Parts

PART I – Application: Classification: Training and Testing (Group Submission)

- a) The notebook must be strictly based on the template provided on Canvas.
- b) You must load the dataset using the [fetch_openml](#) interface provided by scikit-learn, using the following python commands.

```
from sklearn.datasets import fetch_openml  
dataset = fetch_openml(data_id=4538, as_frame=False)
```

- c) You should use at least 4 (but ideally up to 8) classification methods to distinguish between the classes, (see marking scheme). You are encouraged to review the literature and identify methods beyond the ones mentioned in the module. In this case, you should reference the source in your code. Using a method with and without dimensionality reduction counts as two methods; ensure that any variations are listed separately.
- d) You should split the data into training (70%) and testing (30%).
- e) For each classification method, train the model by implementing hyper-parameter tuning of the method.
- f) Evaluate the classification models using appropriate metrics such as
 - Balanced Accuracy,
 - The macro-average and micro-average of the One-vs-Rest ROC AUC score.In addition,
 - Draw the macro-average One-vs-Rest ROCs for all classification methods into a single graph to allow for easy comparison between methods.
 - Draw the micro-average One-vs-Rest ROCs for all classification methods into a single graph to allow for easy comparison between methods.
 - For each class, draw the One-vs-Rest ROCs for all classification methods into a separate graph, to allow for easy comparison between methods.
 - Draw all relevant confusion matrices.
- g) When producing results, use appropriately the print() method or labels for tables and figures, so the meaning of the results is clear without looking at the code.
- h) References (in text) are expected at the end of the notebook.
- i) The Word version of this submission must also include an Appendix on the use of generative AI tools. Provide a full record of prompts and responses used for producing the code, if applicable

PART II – Report (Individual Submission):

This is an individual submission (word or pdf), so you should not collaborate or exchange your views with the other members of your group. The report must strictly follow the structure below, and must not be more than three pages, including text and figures, but excluding references and appendix:

- a. Methods: Briefly list the methods you used and cite the original academic paper, where possible, otherwise cite an appropriate academic source (e.g. textbook). DO NOT explain the methods and DO NOT cite code. In addition, outline the parameters used in your experiments.
- b. Results: Results should be presented in an appropriate way to support your critical arguments and conclusions.
- c. Conclusion: Provide up to three conclusions as a list of bullet points that are directly and logically derived from the discussion above. One of the conclusions must be your recommendation of a single method and parameter configuration. Conclusions such as “the second recommended method is...” or “the worst method is...”) are not acceptable though. Each conclusion should consist of
 - i. a conclusion outline, i.e. a single sentence that uses the following schema: {conclusion statement}, “, as seen in ” {reference to evidence such as table/figure},
 - ii. a short critical discussion (few sentences) to clarify the conclusion and potentially link it to theory.
- b) References: List any references you may have used in your document and any possible use of generative AI, using one of the established referencing systems (e.g. [IEEE](#), [Harvard](#), etc).
- c) Appendix I: Use the template in Appendix I, even if you have not you used generative AI tools for your individual submission. If you used generative AI tools, provide a full record of prompts and responses used for producing this report.
- d) Self-evaluation: Use the template in APPENDIX II to assess both the group submission and your own individual submission, as well as to rate the contributions of the group members towards the group submission.

Marks will be deducted if the report does not follow the proposed structure. Also, any material beyond the page limits will NOT be considered.

Learning outcomes being assessed

- Select and specify suitable methods and algorithms relevant for a particular data analysis process;
- Develop machine learning systems using software packages and/or specialised libraries;
- Articulate and demonstrate the specific problems associated with different phases or tasks of a machine learning pipeline;
- Assess and evaluate machine learning methods using datasets and appropriate criteria.

Marking Scheme (100%): The marking scheme is indicative and further marks may be deducted if the submissions do not follow the assignment guidance. The marking scheme lists the different criteria and when maximum marks will be allocated.

Group Work (40%): The mark for each group member will be weighted according to an estimated average contribution factor, based on all group members' self-assessments. In disputed cases, the teaching team may also consider the students' engagement with the module (attendance, performance on formative quizzes) to adjust the contribution factor.

Criterion	Excellent (8 marks)	Good (6-7 marks)	Satisfactory (4-5 marks)	Below expectations (0-3 marks)
Variety of Methods Attempted (8%)	Uses a wide range of methods (up to 8). Includes several not covered in class. Shows strong initiative.	Uses multiple methods (6~7) including at least one beyond class content.	Uses limited variety of methods (4~5) methods, mainly ones covered methods.	Very few methods (3 or less) attempted.
Code Structure & Documentation (8%)	Code runs correctly; template followed; clear structure. Comments and references identify external and AI-generated content. Appendix I fully clarifies any AI use.	Code runs with minor issues; template mostly followed; comments generally clear and/or Appendix I clarifies any AI use.	Code structure inconsistent; limited or unclear commenting; deviations from template. Lack of clarity about use of AI	Code may not run reliably; lacks structure; missing or unclear references to external/AI content.
Data Usage & Hyperparameter Tuning (8%)	Correct loading preprocessing and splitting. Appropriate training/testing protocol with no data leakage. Wide exploration of parameter values. Results reported for all parameter combinations.	Mostly correct data handling with minor issues. Reasonable parameter exploration	Some weaknesses in preprocessing or splitting; possible leakage concerns. Limited tuning; few parameter values tested.	Incorrect or inappropriate data handling; evidence of data leakage. No meaningful tuning; only default parameters used.
Results (16%)	Complete results for all methods and metrics. Clear well-structured presentation.	Most results included; presentation generally clear.	Some results missing or unclear.	Results incomplete inconsistent or poorly presented.

Individual Work (60%):

Criterion	Excellent (10 marks)	Good (7-9 marks)	Satisfactory (6-7 marks)	Below expectations (0-4 marks)
Structure (10%)	Submission follows expected structure. Self-evaluation is reasonable. Any generative AI use is properly referenced with details in Appendix.	Structure mostly followed; self-evaluation adequate; AI use referenced but minor details missing.	Some structural issues; self-evaluation weak or unclear; AI referencing incomplete.	Poor structure; missing self-evaluation; AI use not referenced.
Methods (10%)	Clear list of methods with proper academic references to original source; parameters clearly provided.	Methods listed with mostly correct references; parameters provided but not fully detailed.	Methods listed but referencing weak; parameter ranges incomplete.	Methods unclear; missing references; parameters not provided.
Results (20%)	Clear presentation using appropriate tables/figures; results easy to interpret.	Results mostly clear; tables/figures adequate.	Some results unclear or poorly formatted.	Results incomplete or difficult to interpret.
Conclusions (20%) Marking scheme assumes three conclusions provided. If fewer conclusions, mark will be adjusted accordingly	Honest and critical discussion; strong insights beyond teaching material; conclusions logically derived and linked to theory; justified recommendation.	Good discussion with some insight; conclusions mostly logical.	Limited insight; conclusions weakly linked to results.	Superficial or unsupported conclusions; little reflection.

Late submissions: for details please refer to the Module Guide and the University Policies.

Plagiarism is a serious academic offence and will always be challenged where it is suspected. When submitting work for assessment ensure it is your own and not someone else's code, ideas, words and images. You must declare that the task submitted for assessment is your own independent work and that all sources used in the submission are referenced.

If generative AI is used, it should be fully acknowledged, as described in the University guidance at [AI and Assessment: Digital Learning & Tools](#). In addition, a full record of prompts and responses received from generative AI tools and used in the submission must be provided. If generative AI is not used, you still need to include the appropriate Appendix in your submissions.

There is a 'right to viva' for this assessment: if the teaching team feels the answers were not authentic to the student, they may arrange for the student to be interviewed and determine the mark for the coursework based on it. In such cases, the teaching team may also consider the students' engagement with the module (attendance, performance on formative quizzes).

Please ensure that your name is on all components of the work presented. For any questions or issues, email: D.Makris@kingston.ac.uk

APPENDIX I – Use of Generative AI

For each submission (group or individual), include the following template at the end.

Delete the sentence that applies to this submission.

- This submission is my/our own independent work and all sources used in the submission are referenced. Generative AI was NOT USED for this submission.
- This submission is my/our own independent work and all sources used in the submission are referenced. Generative AI was USED for this submission, as outlined by the relevant references. For each reference, I/we provide a full record of prompts given to the generative AI tool, and responses received below. I have also included references that acknowledge the usage of AI in the reference list of this coursework, based on the guidance provided at: [AI and Assessment: Digital Learning & Tools](#)

APPENDIX II – Self-Evaluation Template

Include the following template at the end of your individual submission.

For each part, assess the work submitted, by providing a mark from 0-10, according to the marking scheme given in the assignment.

For the group submission (Part I), rate the contribution of each group member (including yourself). Use the following scale:

1. Did not contribute to this criterion
2. Willing but not very successful
3. Average
4. Above Average
5. Outstanding

Access the work submitted			Contribution (1-5 for each group member, including yourself). Replace {member x} with member's name. List members in the same order as at the beginning of the submitted notebook.			
Project Part	Weight	Mark (0-10)	{member 1}	{member 2}	{member 3}	{member 4}
I. Code	4					
II. Report	6					